COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit 3 : Session 2 : SLO 2 — Ripple-Carry vs Carry Look-ahead Adders
Name: JATANPREET SINGH VIRDI JATANPREET SINGH VIRDI | Reg. No: RA2511003011374 | Solved Worksheet

---

Section 1: The Ripple Effect (Understand)

• When you push the first domino it falls and knocks over the next one, which knocks over the one after it, and so on; there is a delay as the fall propagates down the line.

• Similarly, when adding 1111 + 0001, the first full adder generates a carry that must ripple to the second stage, which generates a carry for the third, and so on. The final correct sum is available only after the carry has propagated through all stages — just like waiting for the last domino to fall.

Section 2: Identifying Limitations (Analyze) — single full adder delay = 10 ns

• Total propagation delay for a 4-bit ripple-carry adder: 4 × 10 ns = 40 ns

• Total propagation delay for a 16-bit ripple-carry adder: 16 × 10 ns = 160 ns

• Major limitation: the delay grows linearly with the number of bits. A 64-bit adder would need about 640 ns, making the processor too slow for modern high-speed applications.

Section 3: Introducing a Faster Solution (Remember/Understand)

• A carry look-ahead adder improves performance by reducing the propagation (1) DELAY.

• Instead of waiting for the carry to (2) RIPPLE through each full adder, the CLA uses special logic to calculate all the carry bits (3) SIMULTANEOUSLY.

• It does this by looking at the input bits and determining if a carry will be (4) GENERATED (created at that stage) or (5) PROPAGATED (passed from a previous stage).

• This makes the calculation of each carry (6) INDEPENDENT of the previous carry bit, speeding up the entire addition process.

Section 4: Concept Matching (Analyze)

• 1. Ripple-Carry Adder → B. Carry-out of one stage is the carry-in of the next, creating a chain reaction

• 2. Carry Look-ahead Adder → C. Uses complex logic to calculate all carries in parallel, improving speed

• 3. Propagation Delay → A. Time for an input change to produce a stable output change

• 4. Carry 'Generate' (gi) → E. A carry is created at a bit stage — both input bits are 1

• 5. Carry 'Propagate' (pi) → D. A carry is passed through a bit stage — one input is 1 and the other is 0

Section 5: Justifying the Choice (Evaluate) — high-performance gaming CPU

• Chosen adder: Carry Look-ahead Adder.

• Why better: calculation speed is critical for a gaming CPU. A CLA computes all carry bits in parallel, so it is significantly faster than a ripple-carry adder; arithmetic completes quickly, preventing lag.

• Drawback of the ripple-carry adder: its speed falls as the number of bits grows (e.g., 64-bit calculations) because of the long carry-ripple propagation delay, which would bottleneck the CPU.